Introduction to Artificial Intelligence

Sample educational document for testing a PDF question-answering chatbot.

1. Artificial intelligence

Artificial intelligence (AI) studies systems that perform tasks associated with capabilities such as reasoning, learning, perception, language understanding, and decision-making.

2. Machine learning

Machine learning is a branch of AI in which a system learns patterns from data. Supervised learning uses labeled examples, unsupervised learning seeks structure in unlabeled data, and reinforcement learning learns from rewards and penalties.

3. Training and testing

A dataset is often divided into training, validation, and test sets. Training data fits model parameters, validation data helps select settings, and test data estimates performance on held-out examples.

4. Neural networks

A neural network combines layers of units with adjustable weights. During training, a loss function measures prediction error and an optimization algorithm updates weights to reduce that loss.

5. Generative AI

Generative AI models create content such as text, images, audio, or code. A language model generates text based on patterns learned during training and the context supplied at inference time.

Note: This is a short synthetic sample created for software testing, not an official textbook.